

**Q4: Design the combination circuit with 3 inputs and 1 output, if the sum of inputs is equal to 2**

Using a Karnaugh map:

| BC \ A |    | 3-Variable K-Map (A \ BC) |    |    |    |
|--------|----|---------------------------|----|----|----|
|        |    | 00                        | 01 | 11 | 10 |
| A=0    |    | 0                         | 0  | 1  | 0  |
|        | m0 | m1                        | m3 | m2 |    |
| A=1    |    | 0                         | 1  | 0  | 1  |
|        | m4 | m5                        | m7 | m6 |    |

**Truth Table:**

| A | B | C | F |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

So, The final function is:  $F = A'BC + AB'C + ABC'$



|        |                  |             |              |
|--------|------------------|-------------|--------------|
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| Class: | B2IT/E           | Assignment: | Logic Design |

**Q1: Prove that:  $X' + XY = X' + Y$**

**Solution:**

$$X' + XY + X' = X' + Y \quad (\text{Add } X'Y \text{ to left})$$

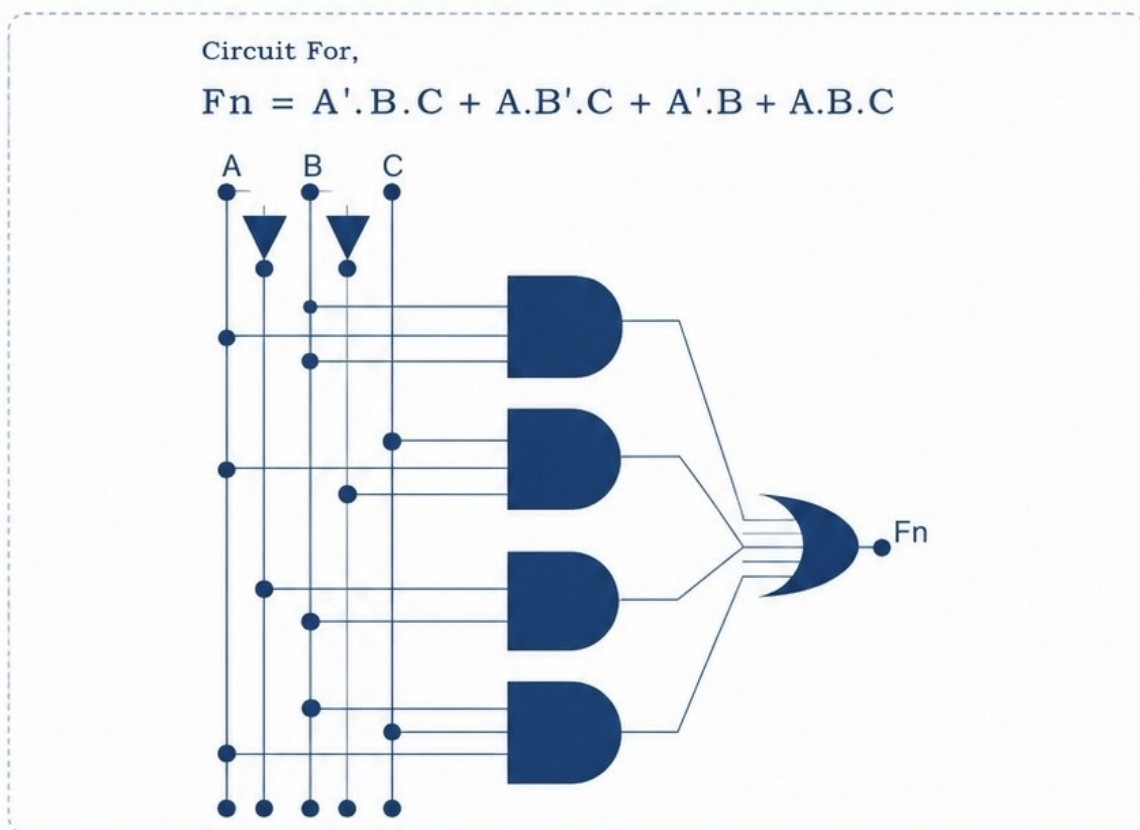
$$X' + Y(X + X') = X' + Y$$

$$X' + Y = X' + Y$$

**Q2: Find the logic circuit for:  $F = A'BC + AB'C + A'B + ABC$**

The Function is:  $A'BC + AB'C + A'B + ABC$

Circuit For,  $F_n = A'.B.C + A.B'.C + A'.B + A.B.C$



### Q3: Simplify the logic function and circuit.

#### K-Map Analysis:

$$F(xyz) = M_0 + M_1 + M_2M_3 + M_4M_6$$

$$\text{Where: } M_2M_3 = 0 \text{ and } M_4M_6 = 0$$

$$\text{Then: } F = M_0 + M_1$$

$$F = x'y'z' + x'y'z$$

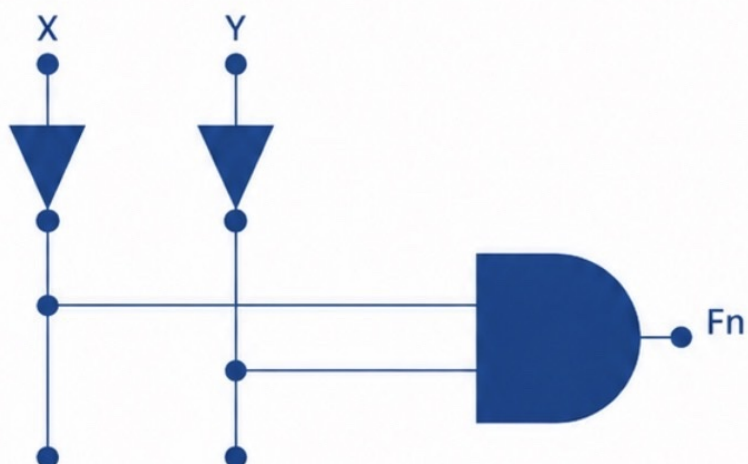
$$F = x'y' (z' + z)$$

$$F = x'y'$$

|   |   | , yz |    |    |    |
|---|---|------|----|----|----|
|   |   | 00   | 01 | 11 | 10 |
| x | 0 | 1    | 1  | 0  | 0  |
|   | 1 | 0    | 0  | 0  | 0  |

Circuit For,  $F_n = X'.Y'$

Circuit For,  
 $F_n = X'.Y'$



**Q5: Find the reduced form for the logic function presented in fig (XYZV)**

**K-Map Analysis:**

|                  | $\bar{z}\bar{e}$ | $\bar{z}e$ | $ze$ | $z\bar{e}$ |
|------------------|------------------|------------|------|------------|
| $\bar{x}\bar{y}$ |                  | 1          |      |            |
| $\bar{x}y$       | 1                | 1          | 1    | 1          |
| $xy$             | 1                | 1          | 1    | 1          |
| $x\bar{y}$       |                  | 1          | 1    |            |

Circuit For,  $F_n = Y + Z'E$

Circuit For,  
 $F_n = Y + Z'E$

